Multiple ergodic averages in mixing systems

One starting point, several observation times.

$$A_N^{\boldsymbol{b}}(x) = \frac{1}{N}\sum_{n=1}^{N}\prod_{j=1}^{d} f_j(T^{b_j n}x)$$

Does the average converge pointwise?
Is the limit the product of the integrals?

Common assumptions and quantifiers

A probability space $(X, \mathcal{F}, \mu)$; an invertible, bimeasurable, measure-preserving T.

$$\mu(E \cap T^{-n}F) \longrightarrow \mu(E)\mu(F) \quad (|n| \to \infty)$$

- Ordinary mixing, for every fixed pair of measurable sets E, F
- Fixed, possibly complex-valued $f_j \in L^\infty(\mu)$
- All integer lengths $N \in \mathbb{N}$; the full-measure set may depend on the functions and slopes

Principal conclusions of the four preprints

On a probability space: invertible, bimeasurable, mixing, measure-preserving T; fixed $f_j \in L^\infty(\mu)$.

D	$d = 3$	$\boldsymbol{b} = (1, 2, 3)$
C	$d = 3$	Distinct nonzero integer slopes; mixed signs allowed
B	$d = 4$	$\boldsymbol{b} = (1, 2, 3, 4)$
A	Any fixed finite $d \geq 2$	$\boldsymbol{b} = (1, \ldots, d)$

$$A_N^{\boldsymbol{b}}(x) \longrightarrow \prod_{j=1}^{d} \int_X f_j \, \mathrm{d}\mu \qquad \text{a.e.}$$

All positive integer N; the full-measure set may depend on the fixed functions and slopes.

Inclusion of conclusions

A	Consecutive integer slopes, any finite order
↓	
A direct corollary of A	Distinct nonzero integer slopes of one sign

Positive slopes: fill unused positions with 1. Negative slopes: apply the argument to T^{-1}.

C covers mixed signs in the triple case. The corollary does not cover mixed signs at arbitrary higher order.

Norm convergence and pointwise convergence

Norm convergence

$$\|A_N - L\|_2 \longrightarrow 0$$

The overall error is small at each length.

Pointwise convergence

$$A_N(x) \longrightarrow L(x) \quad \text{a.e.}$$

The sequence stabilizes at almost every fixed point.

The exceptional points may move with the length. Norm convergence alone does not rule out recurrent bad lengths.

A limit need not be a product

$X = \mathbb{Z}/3\mathbb{Z}$ with uniform measure, $Tx = x + 1$; let $f_1 = f_2 = f_3 = \mathbf{1}_{\{0\}}$.

$$\lim_{N\to\infty} A_N^{(1,2,3)}(x) = \begin{cases} \frac{1}{3}, & x = 0, \\ 0, & x \neq 0, \end{cases} \qquad \prod_{j=1}^{3} \int f_j \, \mathrm{d}\mu = \frac{1}{27}.$$

This system is invertible and ergodic, but not mixing. The example separates the two questions; it does not contradict the mixing claims.

Triple averages: first obtain a limit

1. **Local analysis and variation control**
   Obtain an oscillation estimate uniform over deterministic blocks.
2. **Sublinear growth of oscillation**
   Rule out persistent nonconvergence on a positive-measure set.
3. **Smoothing and length grids**
   Recover all integer lengths; identify the product limit using norm convergence.

Convergence on an auxiliary system does not assert a product limit on every system.

H: the trilinear Hilbert transform

$$T(f_1, f_2, f_3)(x) = \mathrm{p.v.} \int_{\mathbb{R}} f_1(x-t) f_2(x-2t) f_3(x-3t) \, \frac{\mathrm{d}t}{t}$$

$$\|T(f_1, f_2, f_3)\|_{L^1(\mathbb{R})} \leq C \prod_{j=1}^{3} \|f_j\|_{L^3(\mathbb{R})}$$

First stated for Schwartz functions, then uniquely extended as a bounded trilinear operator.

This does not additionally assert pointwise principal-value existence for every L^3 input.

Oscillation requires the local analysis in H and the varying-input control in D.

Higher order: conditional laws after selection

1. **Select bad lengths depending on the point**
   Convert assumed pointwise failure into a positive correlation.
2. **Preserve the needed conditional-law constraints**
   Use auxiliary variables and stabilization of conditional expectations.
3. **Derive a contradiction by extremal selection**
   The selection must preserve the distributional relations used at the end.

Lengths depend on the starting point. Asymptotic factorization at predetermined, well-separated times cannot simply be reused after point-dependent selection.

R: a separate multiple-mixing claim

R claims: ordinary mixing implies mixing of every finite order.

$$\int_X \prod_{j=1}^{s} g_j(T^{t_j}x)\,\mathrm{d}\mu \longrightarrow \prod_{j=1}^{s} \int_X g_j\,\mathrm{d}\mu$$

For fixed finite s and bounded g_j, as $\min_{i\neq j}|t_i - t_j| \to \infty$.

This is a major external claim, not the definition of ordinary mixing and not a time-averaged norm-convergence statement.

Scope and verification status

- A / B: consecutive slopes; A also gives distinct slopes of one sign
- C / D: triple averages; C allows distinct nonzero slopes with mixed signs
- Principal conclusion: almost-everywhere convergence to the product of integrals

No extension here to all unbounded inputs, arbitrary higher-order mixed signs, or quantitative rates.

This report checks fixed-version statements and key dependencies. It does not independently certify the complete proofs in H, R, or the higher-order structural arguments.

Fixed sources and further reading

A Pointwise Multiple Ergodic Averages for Mixing Transformations

B Pointwise convergence of fourfold ergodic averages for mixing transformations

C Triple ergodic averages with distinct integer slopes

D Pointwise convergence of triple ergodic averages for mixing transformations

H An L3 bound for the trilinear Hilbert transform

R Rokhlin's multiple-mixing problem for one transformation

Fixed repository revision: fd4aeeb2ee4f • A–D: 2026-10-04; H: 2026-10-05; R: 2026-09-23.

The main report has 12 slides. The 23 historical technical backup slides have a separate entry.